

Artificial Intelligence and Internet of Things for Robotic Disaster Response

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Abstract— After the Fukushima nuclear disaster and the Wenchuan earthquake, the relevant government agencies recognized the urgency of disaster-straining robots. There are many natural or man-made disasters in Taiwan, and it is usually impossible to dispatch relevant personnel to search or explore immediately. The project proposes to use the architecture of Intelligent Internet of Things (AIoT) (Artificial Intelligence + Internet of Things) to coordinate with ground, surface and aerial and underwater robots, and apply them to disaster response, ground, surface and aerial and underwater swarm robots to collect environmental big data from the disaster site, and then through the Internet of Things. From the field workstation to the cloud for "training" deep learning model and "model verification", the trained deep learning model is transmitted to the field workstation via the Internet of Things, and then transmitted to the ground, surface and aerial and underwater swarm robots for on-site continuing objects classification. Continuously verify the "identification" with the environment and make the best decisions for the response. The related tasks include monitoring, search and rescue of the target.

Index Terms—Artificial Intelligence, Internet of Things, Mobile Robot, Swarm Intelligence, Disaster Response, AIoT

I. INTRODUCTION

Disaster response includes three important tasks: pre-disaster early warning, search and rescue during the disaster, and post-disaster recovery. Disasters include natural (69,227 deaths with economic losses of RMB \$845.1 billion in the Wenchuan earthquake on the mainland China and 15,893 deaths with economic losses US\$210 billion in Great East Japan Earthquake) and man-made (the United States 911 terrorist attack, the death toll is 2,996 people, and the economic loss is US\$178 billion).

In recent years, AI (artificial intelligence) and the Internet of Things (IoT) are emerging trends. The AI and IoT proposed in this paper aimed for disaster response robotic, which construct the IoT architecture connected to existing ground, surface, aerial and underwater robots with national AI research and development infrastructure.

This paper investigates the architecture of swarm intelligence by constructing AIoT architecture for the collaborative operation of heterogeneous robots in disaster response.

AI, Robotics and IoT are attracting widespread attention and are expected to change society's technology in the future. These innovative technologies have the potential to establish (1) a borderless communication society, (2) a symbiotic society between humans and robots, and (3) a safe and reliable

network society. Some specific solutions are proposed [1]: (1) automatic translation solutions, (2) dam detection robot solutions, and (3) large-scale imaging security solutions. All these solutions will collaborate with the cloud and will evolve into an advanced system. In addition, IoT/Edge devices require higher performance and smarter processing, so VLSI plays a more important role.

A vertically integrated four-layer CaSF (cloud-assisted smart factory) architecture is proposed [2]. The key AI technologies involved in CaSF are classified and described according to the logical relationships in the architecture hierarchy. The main problems and technical challenges of AI technology in CaSF system are introduced, and some possible solutions are given.

It's emerging to enhance today's 4G mobile network with fiber and wireless Ethernet technology and AI embedded multi-access edge computing to achieve ultra-reliable and low-latency communication for future immersive haptic Internet applications. Some research ideas are outlined [3], these ideas help to give full play to the full potential of the tactile Internet released by the restrictions imposed by certain vested interests.

A framework for wireless sensor and brake selection in the brake network is proposed [4], which is mainly composed of adaptive neuro-fuzzy inference system. It considers network conditions when selecting roles for different tasks related to network connection restoration.

A middleware is proposed [5] that allows the execution flow of AI applications to be transferred between the device and the cloud. The proposed system aimed to effectively manage device resources and achieve effective communication with various platforms (cloud, mobile).

A smart robot factory is proposed [6], using highly interconnected and deeply integrated smart production lines, and introduced the overall structure, composition, characteristics and advantages of such factories from two aspects of cognitive manufacturing and edge computing.

The cloud service and big data computing infrastructure include National Center for High-performance Computing (NCHC), Taiwan, NVIDIA, Microsoft (Azure), Google and Amazon (AWS).

This paper focuses on a design an AIoT architecture-based swarm intelligence for heterogenous robot collaboration towards the search and rescue tasks for disaster response. It can reduce the casualties of disaster response professionals, and solve the shortage of manpower.

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II. METHOD

A. Scenario

This paper proposes an architecture of AI and IoT in disaster response robotic. Through the IoT and AI algorithms, various unmanned vehicles possess autonomous behavior, and then further becomes the mobile robots. The locomotion proposed in this paper include ground, aerial, surface and underwater. The proposed scenario is illustrated in Figure 1.

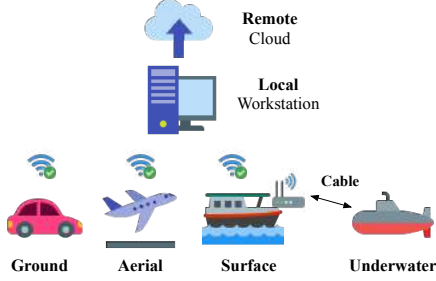


Figure 1. Scenario: Swarm AI for the ground, aerial surface and underwater locomotion.

This paper proposes an intelligent behavior control system for multiple mobile robots as shown in Figure 2. The autonomous robotic behavior includes obstacle avoidance, goal seeking, trajectory tracking and formation. The frontier command is located on the site workstations for collecting raw sensory data and sending task command to robots. Remote AI (strategy) is located in the cloud for highest strategical decision making.

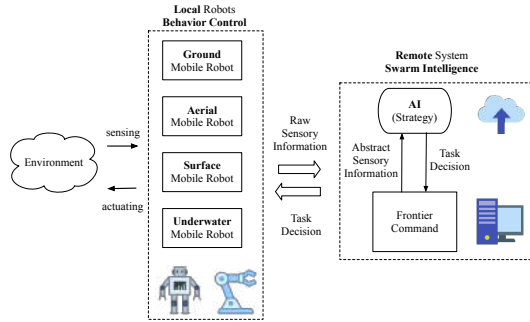


Figure 2. System: Remote swarm AI and local robotic behaviour control.

B. System

The proposed AIoT architecture is illustrated in Figure 3. The workflow is as following:

1. **Data Collection:** sensory data of from environment is transmitted to the cloud through the site workstation.
2. **Training:** Build AI model using the data collected by mobile robots from the environment.
3. **Validation:** evaluation AI model for hyper-parameter tuning and model selection.
4. **Testing:** the trained AI model is further deployed to the local workstation for further testing the

performance. The tested model finally is deployed to the embedded system in mobile robots for continuing learning process.

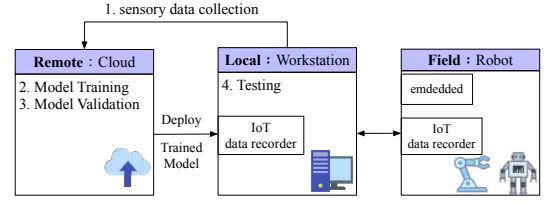


Figure 3. AIoT architecture.

The hierarchical control architectures of autonomous mobile robots proposed in this paper is shown in Figure 4. The high level behavior control system located at the remote end and the low level pose control system located at the mobile robot in the field. The communication includes wireless (distributed) and wire (centralized).

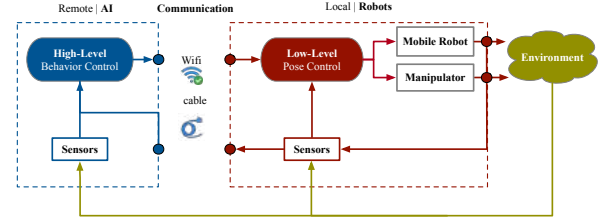


Figure 4. Hierarchical control: high level and low level.

C. Mobile Robot

The locomotion used in the mobile robots are ground, aerial, surface and underwater which is shown in Figure 5, 6, 7, 8 respectively.



Figure 5. Ground locomotion.



Figure 6. Surface locomotion.



Figure 7. Aerial locomotion.



Figure 8. Underwater locomotion.

III. GROUND MODULE

Ground mobile robots have the following advantages:

- The scalability is higher, and the payload and volume limits are much smaller than other types of locomotion, therefore sensors and actuators installed on ground mobile robots can be more complex and sophisticated.
- The disaster is usually occurred on land, e.g. earthquakes, fires, and biochemical disasters.
- The mechanical structure is more mature, e.g. wheel, caterpillar and legged type.

A. Intelligent Robot for Worker Safety Surveillance

The fatal injury rate for the construction industry is higher than the average for all industries. Recently, researchers have shown an increased interest in occupational safety in the construction industry. However, all the current methods using conventional machine learning with stationary cameras suffer from some severe limitations, perceptual aliasing (e.g., different places/objects can appear identical), occlusion (e.g., place/object appearance changes between visits), seasonal / illumination changes, significant viewpoint changes, etc. This work [7] proposes a perception module using end-to-end deep-learning and visual SLAM (Simultaneous Localization and

Mapping) for an effective and efficient object recognition and navigation using a differential-drive mobile robot as shown in Figure 9. The detection result is shown in Figure 10.

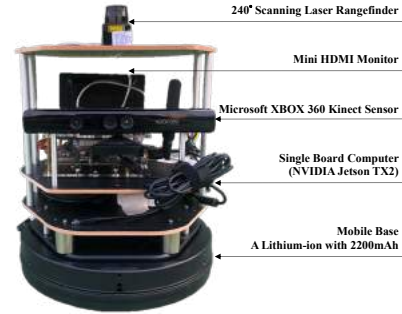


Figure 9. Robot's Configuration.

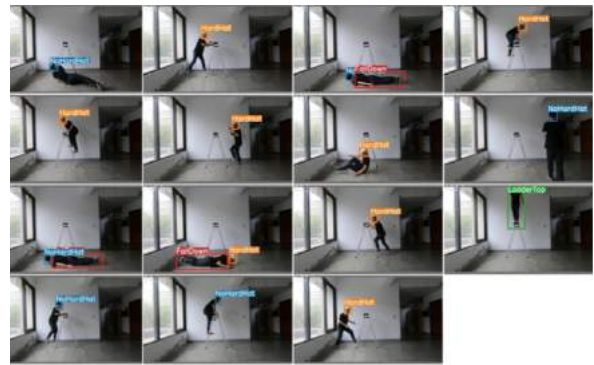


Figure 10. Images of personal protective equipment detection result.

B. Chemical substance detection robot

This work proposes the use of ground mobile robots equipped with micro-spectrometers as shown in Figure 11 for remote chemical content detection [8]. The proposed navigation system is robust under various uncertain factors, and the robot can effectively and efficiently act in various environments. The micro-spectrometer identification system successfully classified various samples. The mobile robot is a two-wheel differential drive mechanism with 8 forward sonars. The command center communicates with the robot through a wireless network. The micro-spectrometer and light source are mounted on the gripper.

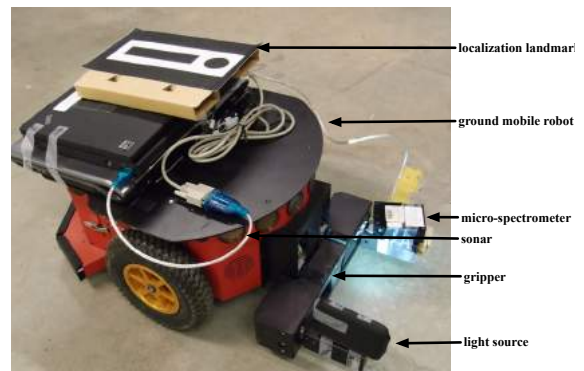


Figure 11. Robot configuration.

Figure 12 shows the motion control, sample gripping, sensing and perception modules for the ground mobile robot.

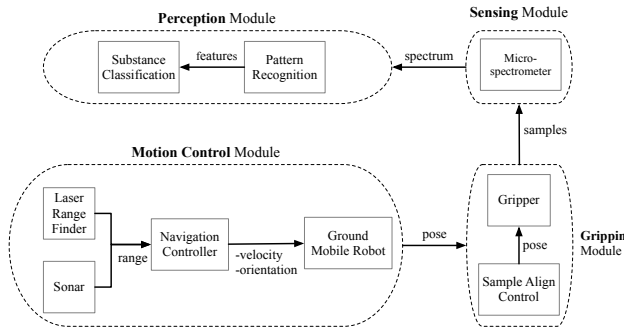


Figure 12. System module.

IV. AERIAL MODULE

Aerial robots are characterized by high speed, flexibility, and wide field of vision. In disaster response, it is used to detect disaster situations, search and rescue trapped people in a wide range, quickly deliver rescue materials, and quickly build disaster environment's models and other tasks.

In this paper, aerial robots are used for face recognition, target measurement, ship search, 3D environment modeling and other functions, as well as ground-aerial, surface-air collaboration based on AI.

A. Hybrid renewable energy power system

This module design a renewable energy system [9] built on the aerial robot, using the sun, rain, air, and lithium batteries to build a sustainable energy system as shown in Figure 13.

Sensing includes solar power, amount of rainfall, oxygen, power available from solar cell, fuel cell and lithium battery, water and hydrogen in the storage.

Power distribution system generate the optimal decision-making according to the sensory data as shown in Figure 14.

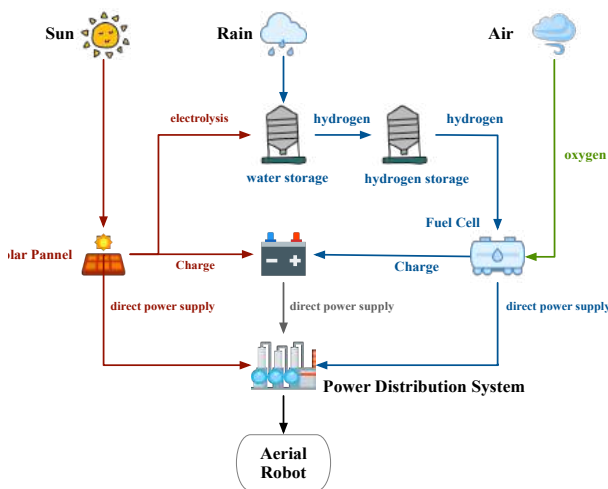


Figure 13. Renewable energy system architecture

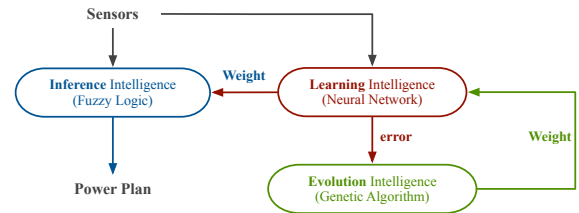


Figure 14. Renewable energy system architecture

B. Face recognition and tracking

During natural and man-made disasters, face recognition can search for missing, injured, elderly, disabilities, and hostages (recognize and locate victims and attackers). This module use aerial robots for face recognition and tracking [10] as shown in Figure 15, to assist search and rescue for shortening the search time and reduce the search area. The workflow is illustrated in Figure 16.



Figure 15. Aerial face recognition and tracking.

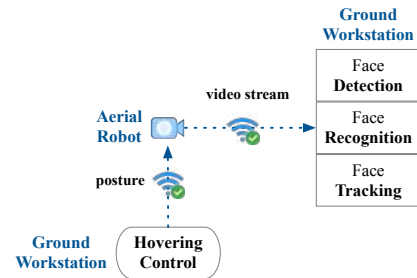


Figure 16. Workflow: aerial face recognition and tracking.

V. SURFACE MODULE

Aerial robot often cannot stay in the air for a long time because of the limited energy carried, and the communication range is limited. Therefore, this module proposes the concept of UAV-based offshore platform.

A. Design and fabrication of surface robot

A surface robot is designed and fabricated to perform forward, backward, and steering movements as shown in Figure 17. Robot with sensors perceive the environment for their position and orientation, find the target, and plan and execute motion. Single propeller generates the propulsion. Propeller rotate clockwise or counterclockwise generate the forward or backward motion respectively. A steering rudder is

connected with a servo motor which control the direction of propeller's propulsion.

The on-board sensor includes sonars, CCD camera, electronic compass, GPS (Global Positioning System), AIS (Automatic Identification System) and IMU (Inertial Measurement Unit). The onboard Wi-Fi station send and receive the control and sensing signal respectively. The actuator includes a motor for controlling propeller (linear speed) and a motor controlling paddle (orientation).

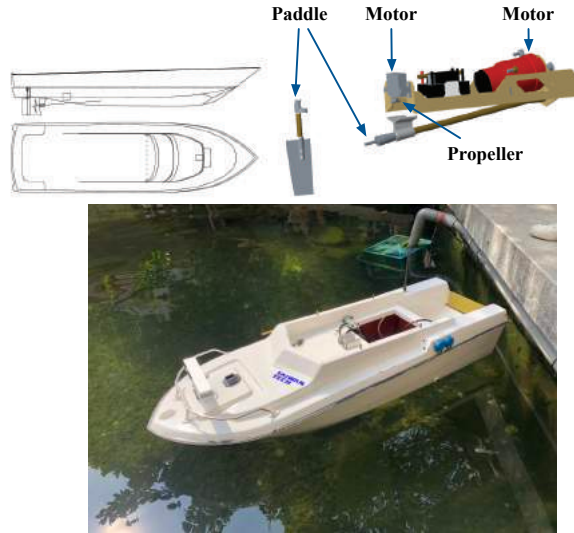


Figure 17. Design and fabrication of unmanned surface vehicle.

B. Surface-Aerial Collaborative Landing

The most critical part of this concept is that UAV can accurately and autonomously land on the ground mobile robot. The fuzzy-neural network and image-based servo control were implemented to complete the landing [11].

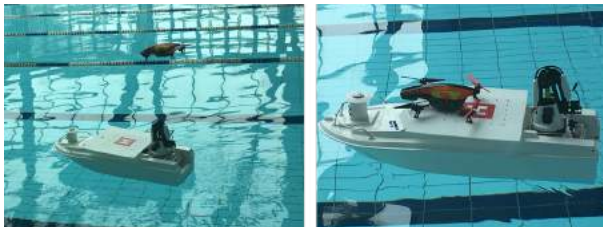


Figure 18. Aerial robot landing on the surface robot.

VI. UNDERWATER MODULE

Underwater vehicle includes (1) Remotely Operated Vehicle (ROV), (2) Autonomous Underwater Vehicle (AUV) and (3) Autonomous Underwater Glider, AUG).

According to the definition, only autonomous underwater vehicles are underwater robots. The system proposed in this paper is aimed at underwater rescue search.

A. Design and fabrication of underwater robot

The design is to control the surfacing, diving and left-right turning by two sets of linearly moving mass. Through the

equivalent center of gravity, to achieve the control of diving and turning of the underwater robot as shown in Figure 19.

The surfacing, diving, left and right turns are controlled by two sets of linear masses, this can effectively reduce the control complexity and save the mass to rotate around the axial circumference to reduce the volume and simplify the shape of the vehicle [12].

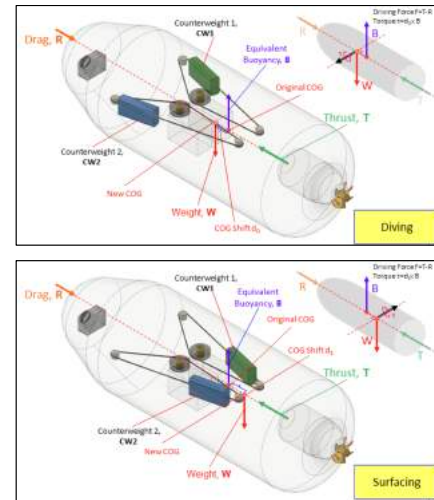


Figure 19. Design of the apparatus for underwater robot motion control with mass shifter mechanism. (Top) diving mode; (Bottom) surfacing mode.

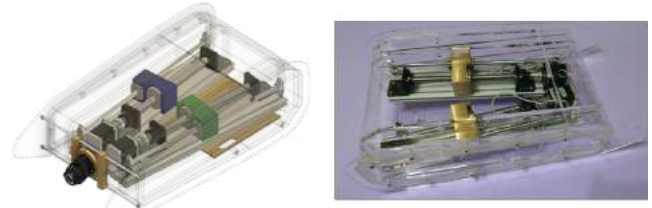


Figure 20. Underwater robot. (Left) CAD drawing; (Right) Actual.

B. Argument Reality for Underwater Object Detection

This module implements a color-correction and augmentation method for underwater object recognition [13] to enhance the recognition rate of models by improving the availability of training data. The underwater robot is used for the underwater images collection as shown in Figure 21. The range measurement based on the vanish point and NN (Neural Network) to build the model are illustrated in Figure 22. The CNN (Convolutional Neural Network) recognition result is shown as in Figure 23.

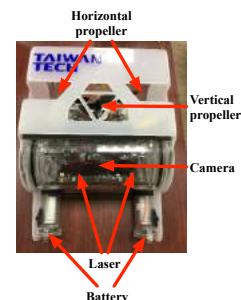


Figure 21. Underwater robot configuration.

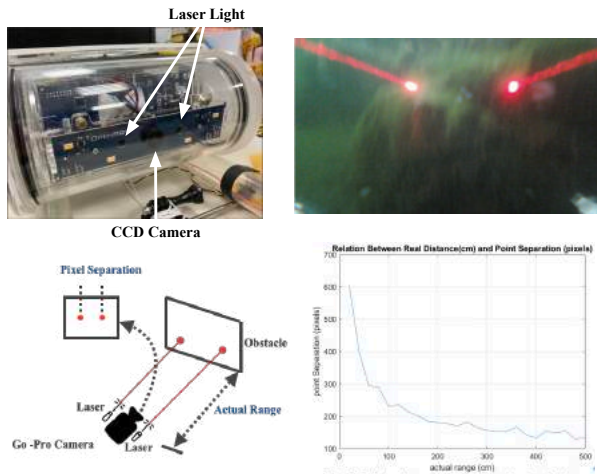


Figure 22. Range measurement.

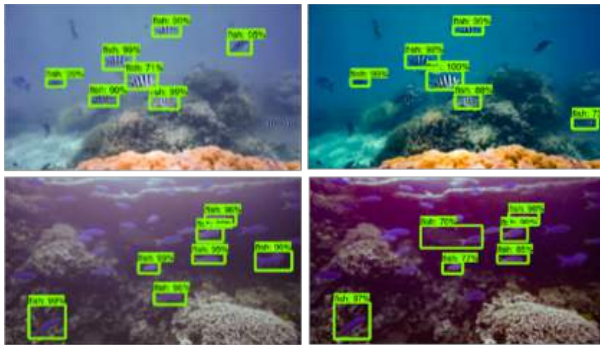


Figure 23. Recognition result.

VII. AI TRAINING MODULE

This work proposes the model which trained on a server deployed to an edge computing device as shown in Figure 24. This module trained and validated deep-learning models on the cloud. It handle each application like storage space, databases, security issues, platforms, architectures etc. AI model can be export and deploy to robots (ground, surface, aerial and underwater) for the object detection or classification tasks.

Edge Computing deployed the AI model to the edge device. It is a distributed computing paradigm. It bring computation and data storage to the specific location without sending data to the cloud server. This way can improve response times and bandwidth. Shorter latency is achieved to reduces transmission costs.

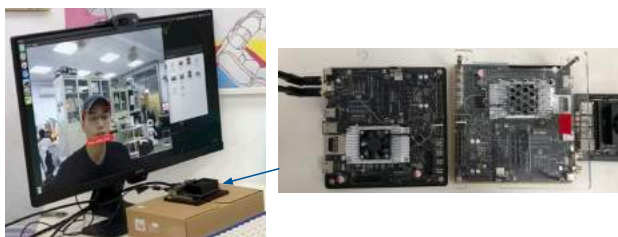


Figure 24. AI training, validation and testing.

VIII. CONCLUSION

This paper proposes four robot cooperation architectures based on Artificial Intelligence and Internet of Things in disaster response for search and rescue. The ground, surface, aerial, and underwater robots individually collect "sensory data" from each's environment and pass to on-site workstations for further cloud computing (training, validation). Trained AI models then deploy back to the on-site workstation via the cloud for testing and further deploy to the embedded system in robots for continuous learning. The results show that the robots cooperate more effectively than the single robot. The results also show that decision-making using artificial intelligence is more in line with the disaster response environment. The research focus of this paper is to improve the limitations of a single robot in a disaster response environment. Therefore, the swarm intelligence among heterogenous robots is proposed.

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